Curriculum Vitae

Dr. Wooyeon Jo

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Education and Research History

- *July. 2022 – Current* : Postdoctoral Research Associate, Virginia Commonwealth University
  PI: Prof. Irfan Ahmed
  Major: Digital Forensics and Threat Intelligence
- *Mar. 2015 – Feb. 2022* : Ph.D. in Computer Engineering, Ajou University ICS Lab
  (Integrated Ph.D. Program)
  PI: Prof. Taeshik Shon
  Major: Digital Forensics and Cyber Security
- *Mar. 2011 – Feb. 2015* : B.S. in Computer Engineering, Ajou University
  Major: Computer Engineering

Honors

- USCYBERCOM **Hunter** Award for MEMWAR at Cyber RECon 2026, presented by Rear Admiral Kevin Lenox, the Chief of Staff of U.S. Cyber Command. The award ceremony was organized at Defense Information School, Fort Meade, MD, April 2026
  - With a congratulatory letter from the President of Virginia Commonwealth University.
- USCYBERCOM **Guardian** Award for 3-Channel Validator at Cyber RECon 2025, presented by Mike Clark, the Deputy Director of Plans and Policy (J5) of U.S. Cyber Command. The award ceremony was organized at National Cryptologic Museum, Annapolis Junction, MD, February 2025
- USCYBERCOM **Guardian** Award and **Commander's Cup** (Best in Show) for NAZAR at Cyber RECon 2024, among teams from 17 schools (as First Author), presented by Four-star General Timothy D. Haugh, then Commander of U.S. Cyber Command. The award ceremony was organized at US Naval Academy, Annapolis, MD, April 2024
  - With USCYBERCOM Challenge Coin awarded by Four-star General Timothy D. Haugh.
  - With USCYBERCOM Challenge Coin awarded by Rear Admiral Dennis Velez, the director of plans and policy of United States Cyber Command
- Best Student Paper Award, 23rd Annual Digital Forensics Research Conference (DFRWS

US’23), hosted at Johns Hopkins University, Baltimore, MD, July 2023

- Graduate Student Representative, February 2022
- Student Scholarship from DFRWS USA 2019
- Future Computing Scholarship 2014
- Information Computer Scholarship 2013

Graduated Theses

- Doctor of Philosophy
  - Smart Digital Forensics:
    An Application of Supervised Learning to Industrial Network and System
  - Committee: Dr. Manpyo Hong(Chair), Dr. Taeshik Shon, Dr. Jin Kwak, Dr. Tae-Sun Chung(Ajou University), Dr. Changhoon Lee(Seoul National University of Science & Technology)

Research Interests

- Digital Forensics for Industrial and Embedded Systems
  - PLC Memory Acquisition and Analysis
  - Reverse Engineering of Proprietary Protocols and Memory Layouts
  - File Systems, Volatile Memory, and Smart Devices
- Industrial Control System Security
  - Design-Level Vulnerability Discovery and Coordinated Disclosure
  - Control Logic Attacks, Firmware and Logic Integrity
  - Intrusion and Anomaly Detection for Industrial and In-Vehicle Networks
- Machine Learning for Security and Forensics
  - Automated Analysis of Undocumented Device Memory
  - Forensic Triage of Large-Scale ICS Data

Publications

International Journals (SCI/E)

- S Kim, **Wooyeon Jo**, et al. "Two-Phase industrial control system anomaly detection using communication patterns and deep learning." *Electronics* 13.8 (2024): 1520.
- M Kim, Y Shin, **W Jo**, T Shon. "Digital forensic analysis of intelligent and smart IoT devices." *The Journal of Supercomputing* 79.1 (2023): 973-997.
- **Wooyeon Jo**, et al. "Automatic Whitelist Generation System for Ethernet based In-Vehicle Network." *Computers in Industry* 142 (2022): 103735.
- Y Shin, S Kim, **W Jo**, T Shon. "Digital forensic case studies for in-vehicle infotainment systems using Android Auto and Apple CarPlay." *Sensors* 22.19 (2022): 7196.
- H Kim, Y Shin, S Kim, **W Jo**, M Kim, T Shon. "Digital forensic analysis to improve user privacy on Android." *Sensors* 22.11 (2022): 3971.
- H Kim, S Kim, Y Shin, **W Jo**, S Lee, T Shon. "Ext4 and XFS file system forensic framework based on TSK." *Electronics* 10.18 (2021): 2310.
- H Kim, S Kim, **W Jo**, K Kim, T Shon. "Unknown Payload Anomaly Detection Based on Format and Field Semantics Inference in Cyber-Physical Infrastructure Systems." *IEEE Access* 9 (2021): 75542-75552.
- **Wooyeon Jo**, et al. "Packet Preprocessing in CNN-Based Network Intrusion Detection System." *Electronics* 9.7 (2020): 1151.
- Lee, Seokjun, **Wooyeon Jo**, Soowoong Eo, and Taeshik Shon. "ExtSFR: scalable file recovery framework based on an Ext file system." *Multimedia Tools and Applications* 79 (2020): 16093-16111.
- Kim, SungJin, **Wooyeon Jo**, and Taeshik Shon. "APAD: autoencoder-based payload anomaly detection for industrial IoE." *Applied Soft Computing* 88 (2020): 106017.
- Kim, Myungjong, **Wooyeon Jo**, and Taeshik Shon. "Visualization for internet of things: power system and financial network cases." *Multimedia Tools and Applications* 78.3 (2019): 3241-3265.
- **Wooyeon Jo**, Hyunsoo Chang, and Taeshik Shon. "Digital forensic science approach by file recovery research." *The Journal of Supercomputing* 74.8 (2018): 3704-3725.

International Conference

- **Wooyeon Jo,** and Irfan Ahmed. "Oops, It Halted Again: Exploiting PLC Memory for Fun and Profit in Industrial Control Systems." 19th USENIX WOOT Conference on Offensive Technologies (WOOT 25). 2025, pp. 191-207.
  - *With CVE-2024-11737 (CISA ICS Advisory ICSA-24-352-04), CVSS v3.1 9.8 / v4.0 9.3, Critical*

- Ayub, Adeen, **Wooyeon Jo**, and Irfan Ahmed. "Charlie, Charlie, Charlie on Industrial Control Systems: PLC Control Logic Attacks by Design, Not by Chance." 2024 IEEE International Symposium on Hardware Oriented Security and Trust (HOST). IEEE, 2024, pp. 182-193.

- [Magazine] A Ayub, **W Jo**, SA Qasim, I Ahmed. "How are industrial control systems insecure by design? a deeper insight into real-world programmable logic controllers." *IEEE Security & Privacy* 21.4 (2023): 10-19.

- A Ayub, N Zubair, H Yoo, **W Jo**, I Ahmed. "Gadgets of gadgets in industrial control systems: Return oriented programming attacks on PLCs." 2023 IEEE International Symposium on Hardware Oriented Security and Trust (HOST). IEEE, 2023, pp. 215-226.

- Qasim, Syed Ali, **Wooyeon Jo**, and Irfan Ahmed. "PREE: Heuristic builder for reverse engineering of network protocols in industrial control systems." DFRWS USA 2023. Published in Forensic Science International: Digital Investigation 45 (2023): 301565.
  - ***(Best Student Paper Award)***

- M Kim, Y Shin, **W Jo**, T Shon. "Security analysis of smart watch and band devices." 2021 International Conference on Computational Science and Computational Intelligence (CSCI). IEEE, 2021, pp. 655-658.

- Y Shin, H Kim, S Kim, D Yoo, **W Jo**, T Shon. "Certificate Injection-Based Encrypted Traffic Forensics in AI Speaker Ecosystem." DFRWS USA 2020. Published in Forensic Science International: Digital Investigation 33 (2020): 301010.

- **Wooyeon Jo**, et al. "Digital forensic practices and methodologies for AI speaker ecosystems." DFRWS USA 2019. Published in Digital Investigation 29 (2019): S80-S93.

- Kim, Hyungchan, **Wooyeon Jo**, et al. "Digital Forensic Analysis using Android Application Cache Data." 2019 International Conference on Platform Technology and Service (PlatCon). IEEE, 2019.

- Shin, Yeonghun, **Wooyeon Jo**, et al. "An Security Analysis of Ext Filesystem metadata." 2019 4th Technology Innovation Management and Engineering Science International Conference (TIMES-iCON). IEEE, 2019.

- Jeong, Jaehan, **Wooyeon Jo**, et al. "Backtracking and Visualization Techniques for Financial Network Security." 2018 IEEE International Conference of Safety Produce Informatization (IICSPI). IEEE, 2018.

- Kim, Hyunjin, **Wooyeon Jo**, et al. "A novel security framework for industrial iot based on isa 100.11 a." International conference on heterogeneous networking for quality, reliability, security and robustness. Springer, Cham, 2018.

- **Wooyeon Jo**, Hyunsoo Chang, and Taeshik Shon. "Digital forensic approach for file recovery in Unix systems: Research of data recovery on Unix file system." 2016 IEEE Information Technology, Networking, Electronic and Automation Control Conference. IEEE, 2016, pp. 562-565.

- Kim, SungJin, **Wooyeon Jo**, and Taeshik Shon. "A novel vulnerability analysis approach to generate fuzzing test case in industrial control systems." 2016 IEEE Information Technology, Networking, Electronic and Automation Control Conference. IEEE, 2016, pp. 566-570.

- Eo, Soowoong, **Wooyeon Jo**, et al. "A phase of deleted file recovery for digital forensics research in Tizen." 2015 5th International Conference on IT Convergence and Security (ICITCS). IEEE, 2015.

USCYBERCOM Cyber RECon (Reviewed by U.S. Cyber Command)

- **Wooyeon Jo**, Badreddin Alhabbal, Muhammad Shaharyar Yaqub, Nehal Ameen, Yahya Forihat, and Irfan Ahmed. "MEMWAR: Controller Memory Exploitation and Weaponization in Critical Infrastructure." USCYBERCOM's Cyber RECon 2026.
  - ***USCYBERCOM Hunter Award, 2026***

- Muhammad Ahsan, **Wooyeon Jo**, Muhammad Shaharyar Yaqub, Nehal Ameen, Yahya Forihat, Hala Ali, and Irfan Ahmed. "3-Channel Validator: A Multi-Layered Framework for Anomalous PLC Detection in SCADA Systems." USCYBERCOM's Cyber RECon 2025.
  - ***USCYBERCOM Guardian Award, 2025***

- **Wooyeon Jo**, Adeen Ayub, Muhammad Ahsan, Nehal Ameen, and Irfan Ahmed. "NAZAR: Stateful Intrusion Detection System for Control Logic Attacks in Industrial Control Systems." USCYBERCOM's Cyber RECon 2024.

- ***USCYBERCOM Guardian Award and Commander's Cup (Best in Show), 2024 (link)***

Domestic Journal (Korea)

- **Wooyeon Jo**, M Kim, K Park, M Hong, J Kwak, and T Shon. "Anomaly detection using visualization-based network forensics." *Journal of the Korea Institute of Information Security & Cryptology* 27.1 (2017): 25-38.

Patents(Korea)

- [Granted] METHOD AND APPARATUS FOR VISUALIZING ANOMALY DETECTION IN NETWORK FORENSICS, 10-1976395, 2019.05.02, doi:10.8080/1020170022915
- [Granted] METHOD FOR MANAGING DATA OF FILE SYSTEM AND COMPUTING SYSTEM USING THE SAME, 10-1836380, 2018.03.02, doi:10.8080/1020160063614
- [Granted] METHOD FOR MANAGING DATA OF FILE SYSTEM AND COMPUTING SYSTEM USING THE SAME, 10-1769778, 2017.08.14, doi:10.8080/1020160055566
- [Granted] APPARATUS AND METHOD FOR DIGITAL FORENSIC FOR IOT SYSTEM BASED ON CLOUD, 10-2156174, 2020.09.09, doi:10.8080/1020190075441

Programming Skills

- Expert in Python/C/JAVA programming with network-related project experiences.
- Experienced in programming with Web with network security visualization project experiences.
- Experienced in programming with C++ on Windows, Linux and Android NDK environments.

Teaching Experience

As a graduate student in the doctoral program, I took one or more TAs every semester. The following describes a case where I gained teaching experience by taking lab classes for the entire semester during the course. I have provided these hands-on lectures to not only undergraduate students of Ajou University, but also military personnel and gifted education.

- **Guest Lecturer,** Virginia Commonwealth University
  - Spring 2023, ICS Security and Attacks
  - Spring 2024, ICS Security and Attacks
  - Fall 2025, Memory Forensics with Volatility 3
  - Network Security (undergraduate course)
  - Summer 2026, Research 101, NSF Research Experiences for Undergraduates (REU)

Summer Program

- **Workshop Instructor,** SleuthCPS 2025 Workshop, sponsored by the Commonwealth Cyber Initiative (CCI)
  - Built the training platform and taught the lecture and hands-on lab on PLC memory forensics
  - Designed a capture-the-flag challenge set in a realistic ICS environment

- **Teaching Assistant,** Ajou University
  - Fall 2017, Software Application Security
  - Fall 2018, Software Application Security
  - Capstone Design (senior capstone course), evaluated team presentations and final deliverables
  - Led the lab sessions of undergraduate and graduate security courses for four years, with eight hands-on labs and four projects (30 to 40 students per section)
  - Led several lectures on Digital Forensics (file system, memory, automotive, mobile, and smart devices), Network Security, and ICS on behalf of the professor

Professional Service

- **Reviewer** (2026): DFRWS USA, DFRWS APAC
- **Sub-reviewer** (2026): IEEE HOST, ACSAC, ACM CODASPY, NDSS
- **Co-organizer**, SleuthCPS 2025 Workshop, sponsored by the Commonwealth Cyber Initiative (CCI)

Projects

- File system forensic analysis tool development, Supreme Prosecutors' Office Republic of Korea, 2012.10.01 ~ 2017.08.31.
- BK21 Plus, National Research Foundation of Korea, 2013.09 ~ 2020.08
- SG heterogeneous protocol gateway security module development, KEPCO(Korea Electric Power Corporation) KDN (Knowledge, Data and Network), 2014.12 ~ 2015.05
- A Study on Vulnerability Analysis Technology in Implementation of Control Protocol, National Security Research Institute, 2015.04 ~ 2015.11
- Development of digital evidence analysis and restoration software based on open platform file system in IoT environment, Ajou University LINK Project Team, 2015.07 ~ 2016.01
- Control system network forensic architecture study, National Security Research Institute, 2016.4. ~ 2016.10.
- Industrial IoT target vulnerability fuzzing and digital forensic technology development, Ajou University LINK Project Team, 2016.07. ~ 2017.01.
- ISA 100.11a based power control system wireless security technology development, KEPCO(Korea Electric Power Corporation) Research Institute, 2016.12 ~ 2017.12.
- Industrial wireless communication protocol security threat analysis study, National Security Research Institute, 2018.04 ~ 2018.10
- Preemptive recognition of manufacturing process abnormalities to solve the problem of smart factory shutdown due to cyber attacks, Institute for Information & Communication Technology Planning & Evaluation (IITP), 2018.04 ~ 2020.12
- Development of digital forensic integrated platform, Institute for Information & Communication Technology Planning & Evaluation (IITP), 2018.05 ~ 2020.12
- A study on the storage data system of cloud server-based artificial intelligence assistants and devices, Supreme Prosecutors' Office Republic of Korea, 2018.06 ~ 2018.12
- Advancement of power control system abnormal behavior detection algorithm and research on detection equipment development, KPX(Korea Power Exchange), 2018.07 ~ 2020.07
- AI Speaker Forensic Analysis, Korea Institute of Information Security & Cryptology, 2019.03~2019.11
- Deep Learning-based Abnormal Behavior Detection Technology Research in Vehicle Ethernet Environment, HYUNDAI Research Project for Future Technology, 2019.08~2020.02
- A study on how to completely delete data in Android OS (9,10), National Security Research Institute, 2020.04 ~ 2020.10
- A study on analysis techniques for wearable devices (smart watches and bands). Korean National Police Agency, 2020.09 ~ 2020.12

- Reliable intelligent threat detection engine research for national infrastructure cybersecurity, National Security Research Institute, 2018.06 ~ 2023.02
- Vehicle Infotainment Forensic Analysis, Korea Institute of Information Security & Cryptology, 2020.03 ~ 2021.11
- Ethernet Communication Diagnostics, HYUNDAI, 2021.06 ~ 2022.06
- Funded by the U.S. Department of Homeland Security under Grant Award Number 17STCIN00001-05-00, 2022.09 ~ Current

Professional References

- Dr. Irfan Ahmed, Professor, Virginia Commonwealth University, iahmed3@vcu.edu
- Dr. Heng Yin, Professor, University of California, Riverside, heng@cs.ucr.edu
- Dr. Taeshik Shon, Professor, Ajou University, tsshon@ajou.ac.kr
- Dr. Adeen Ayub, Assistant Professor, James Madison University, ayubax@jmu.edu

Personal References available upon request

I hereby declare that the above information is true and correct in every respect to the best of my knowledge.

11/2026

Wooyeon Jo